

A Practical Guide to Variational autoencoder

TOPIC -- VARIATIONAL AUTOENCODER

$$\mathcal{L} = \mathbb{E}_{q_\phi}[\log p_\theta(x|z)] - D_{KL}(q_\phi(z|x) \| p(z))$$

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where $p_\theta(x, z)$ represents the joint distribution under p_θ of the observable data x and its latent representation or encoding z . According to the chain rule, the equation can be rewritten as

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Statistical distance VAE variants After the initial work of Diederik P. Kingma and Max Welling, several procedures were proposed to formulate in a more abstract way the operation of the VAE. In these approaches the loss function is composed of two parts :

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$$\begin{aligned} D_{KL}(q_\phi(z|x) \| p_\theta(z|x)) &= \mathbb{E}_{z \sim q_\phi(\cdot|x)} [\ln \frac{q_\phi(z|x)}{p_\theta(z|x)}] \\ &= \mathbb{E}_{z \sim q_\phi(\cdot|x)} [\ln \frac{q_\phi(z|x)}{p_\theta(x)} \frac{p_\theta(x)}{p_\theta(x,z)}] = \ln \frac{p_\theta(x)}{p_\theta(x,z)} + \mathbb{E}_{z \sim q_\phi(\cdot|x)} [\ln \frac{q_\phi(z|x)}{p_\theta(x,z)}] . \end{aligned}$$

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In the vanilla variational autoencoder, z is usually taken to be a finite-dimensional vector of real numbers, and $p_\theta(x|z)$ to be a Gaussian distribution. Then $p_\theta(x)$ is a mixture of Gaussian distributions. It is now possible to define the set of the relationships between the input data and its latent representation as

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β -VAE is an implementation with a weighted Kullback–Leibler divergence term to automatically discover and interpret factorised latent representations. With this implementation, it is possible to force manifold disentanglement for β values greater than one. This architecture can discover disentangled latent factors without supervision. The conditional VAE (CVAE), inserts label information in the latent space to force a deterministic constrained representation of the learned data. Some structures directly deal with the quality of the generated samples or implement more than one latent space to further improve the representation learning. Some architectures mix VAE and generative adversarial networks to obtain hybrid models. It is not necessary to use gradients to update the encoder. In fact, the encoder is not necessary for the generative model.

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Formulation From the point of view of probabilistic modeling, one wants to maximize the likelihood of the data x by their chosen parameterized probability distribution $p_{\theta}(x) = p(x|\theta)$. This distribution is usually chosen to be a Gaussian $N(x|\mu, \sigma)$ which is parameterized by μ and σ respectively, and as a member of the exponential family it is easy to work with as a noise distribution. Simple distributions are easy enough to maximize, however distributions where a prior is assumed over the latents z results in intractable integrals. Let us find $p_{\theta}(x)$ via marginalizing over z .

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To efficiently search for $\theta^*, \phi^* = \arg\max_{\theta, \phi} L_{\theta, \phi}(x)$ the typical method is gradient ascent. It is straightforward to find $\nabla_{\theta} E_{z \sim q_{\phi}(\cdot|x)} [\ln p_{\theta}(x, z) q_{\phi}(z|x)] = E_{z \sim q_{\phi}(\cdot|x)} [\nabla_{\theta} \ln p_{\theta}(x, z) q_{\phi}(z|x)]$. However, $\nabla_{\phi} E_{z \sim q_{\phi}(\cdot|x)} [\ln p_{\theta}(x, z) q_{\phi}(z|x)]$ does not allow one to put the ∇_{ϕ} inside the expectation, since ϕ appears in the probability distribution itself. The reparameterization trick (also

known as stochastic backpropagation) bypasses this difficulty. The most important example is when $z \sim q_\phi(\cdot | x)$ is normally distributed, as $N(\mu_\phi(x), \Sigma_\phi(x))$.

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This can be reparametrized by letting $\epsilon \sim N(0, I)$ be a "standard random number generator", and construct z as $z = \mu_\phi(x) + L_\phi(x) \epsilon$. Here, $L_\phi(x)$ is obtained by the Cholesky decomposition: $\Sigma_\phi(x) = L_\phi(x) L_\phi(x)^T$. Then we have $\nabla_\phi E z \sim q_\phi(\cdot | x) [\ln p_\theta(x, z) q_\phi(z | x)] = E [\nabla_\phi \ln p_\theta(x, \mu_\phi(x) + L_\phi(x) \epsilon) q_\phi(\mu_\phi(x) + L_\phi(x) \epsilon | x)]$
$$\nabla_\phi \mathbb{E} [z \sim q_\phi(\cdot | x)] \ln \left(\frac{p_\theta(x, z)}{q_\phi(z | x)} \right) = \mathbb{E} [\nabla_\phi \ln \left(\frac{p_\theta(x, \mu_\phi(x) + L_\phi(x) \epsilon)}{q_\phi(\mu_\phi(x) + L_\phi(x) \epsilon | x)} \right)]$$
 and so we obtained an unbiased estimator of the gradient, allowing stochastic gradient descent. Since we reparametrized z , we need to find $q_\phi(z | x)$. Let q_0 be the probability density function for ϵ , then $\ln q_\phi(z | x) = \ln q_0(\epsilon) - \ln |\det(\partial \epsilon / \partial z)|$ where $\partial \epsilon / \partial z$ is the Jacobian matrix of z with respect to ϵ . Since $z = \mu_\phi(x) + L_\phi(x) \epsilon$, this is $\ln q_\phi(z | x) = -\frac{1}{2} \epsilon^T \epsilon - \ln |\det L_\phi(x)| - \frac{n}{2} \ln(2\pi)$
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